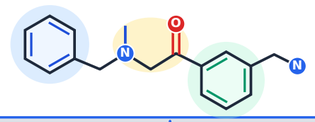
ArcMDTI Architecture: Structure-Informed Cross-Attention & Hyperspherical Margin Learning

STAGE 1 — Biomolecular Input

SMILES & 3-State Structural Priors

(a) Drug Chemical Stream

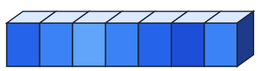
Small-Molecule Ligand (SMILES)



Chemical Substructure Tokens

C | 1 | = | N | C | C | (=O) | C | ···

3D Drug Embedding Tensor



100 × 128

(b) Structure-Informed Protein Stream

Primary Sequence

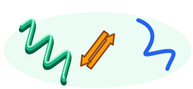
Amino Acid Chain

M A D K T G V

Residue Token Track

Secondary Structure

NetSurfP-3.0 Motifs



Helix (H) · Sheet (E) · Coil (C)

(+)

3D Structure-Informed Protein Tensor



1000 × 128

STAGE 2 — Multi-Scale Encoder

CNN – SE – BiLSTM – Transformer

Drug Multi-Scale Hybrid Encoder

Multi-Scale Dilated CNN

$k=3$ Local | $k=5, d=2$ Dilated

SE Channel Attention

Dynamic channel weighting

BiLSTM Context

Bidirectional sequence flow

Transformer Encoder

Global Multi-Head Self-Attn

Protein Multi-Scale Hybrid Encoder

Multi-Scale Dilated CNN

$k=3$ Local | $k=5, d=2$ Dilated

SE Channel Attention

Dynamic channel weighting

BiLSTM Context

Bidirectional sequence flow

Transformer Encoder

Global Multi-Head Self-Attn

Local Pharmacophore → Channel Recalibration → Global Context

STAGE 3 — Cross-Modal Fusion

Bidirectional Attention & Dynamic Gate

Drug Contextual Tensor

Protein Contextual Tensor

Bidirectional Multi-Head Cross-Attention Bridge

Drug → Protein

Substructure Query

Q_d K_p V_p

Attention Affinity

Pocket-Ligand Grid

Protein → Drug

Pocket Residue Query

Q_p K_d V_d

Dual-Statistic Pooling

Mean Pool (Global Profile)

Max Pool (Salient Motif)

Masked Dual Pooling

Masked Mean (Domain)

Masked Max (Pocket Peak)

Joint Multi-Modal Representation (512-D)

Dynamic Sigmoidal Pocket Gate

Identity Stream

Sigmoid Gate $\sigma(\cdot)$

Noise suppression

Gated Pocket Tensor (512-D)

STAGE 4 — Hyperspherical Head

Geodesic Margin & Focal Modulation

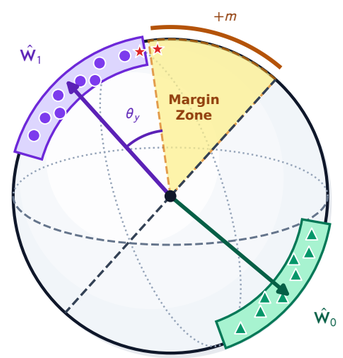
Deep Projector & Unit Normalization

MLP Projector (512-D → 256-D)

Unit $\hat{\mathbf{h}} \in \mathbb{S}^{255}$ ($\|\hat{\mathbf{h}}\|_2 = 1$)

3D Unit Hypersphere ($\mathbb{S}^{255}$) Margin Space

ArcFace Geodesic Penalty + Focal Hard-Example Mining



Active Cap ($y=1$) | Inactive Cap ($y=0$)

Hard Boundary Pairs (Amplified by Focal Weight)

Margin Logit: $s \cdot \cos(\theta_y + m)$

Calibrated DTI Interaction Output

Active Binding (P = 0.94)

Key Visual Components: 3D Tensor Cubes = Multi-Channel Feature Maps; (+) = Element-Wise / Residual Addition; (⊙) = Hadamard Gating; ★ = Focal-Weighted Hard Boundary Pairs; +m = Geodesic Angular Margin.